



COMPUTER AWARENESS

1. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0- 16383) and each cylinder contains 64 sectors (0-63). Data are organized cylinder- wise and the addressing format is < cylinder no. , surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9,40>. What is the cylinder number of the last sector of the file, if it is stored in a continuous manner?

- (a)1284 (b)1282
(c)1286 (d)1288

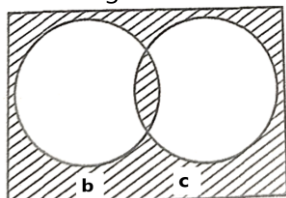
2. Consider the following minterm expression for F.

$$F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$$

The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimum sum- of- product form for F is

- (a) $Q\bar{S} + \bar{Q}S$
(b) $QS + \bar{Q}\bar{S}$
(c) $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$
(d) $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

3. The boolean expression represented by the following Venn diagram is



- (a) a XOR b (b) $a'b + ab$
(c) $ab + a'b$ (d) $(a + b')(a' + b)$

4. The range of n-bit signed magnitude representation is

- (a) 0 to $2^n - 1$
(b) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
(c) $-(2^n - 1)$ to $(2^{n-1} - 1)$
(d) 0 to $2^{n-1} - 1$

5. A hard disk has a rotational speed of 6000 rpm. Its average latency time is

- (a) 5×10^{-3} sec (b) 00.05 sec
(c) 1 sec (d) 0.5 sec

6. The 2's complement representation of the number $(-100)_{10}$ in an 8-bit computer is

- (a) 10011011 (b) 01100100
(c) 11100100 (d) 10011100

7. The number of terms in the product of sum canonical form of $|(x_1 + x_2)(x_3x_4)|$ is

- (a) 7 (b) 8
(c) 9 (d) none of these

8. Find the odd out:

- (a) HTTP (b) FCFS
(c) HTML (d) TCP/IP

9. Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solution are

- (a) 4 (b) 6
(c) 5 (d) 7

10. Subtract $(1010)_2$ from $(1101)_2$ using first complement

- (a) $(1100)_2$ (b) $(0101)_2$
(c) $(1001)_2$ (d) $(0011)_2$

ANSWER

1.	A	2.	B	3.	C	4.	B	5.	A
6.	D	7.	*	8.	B	9.	C	10.	d

SOLUTION

1. (a)

42797 KB will take 85512 sectors ($42797 \times 1024 \text{ bytes} / 512 \text{ bytes}$)
Since, there are 64 sectors per surface $85512 / 64 = 1337.406$ sectors are required, so we take 1338 sectors. These sectors are distributed among 16 surfaces, so $1338 / 16 = 83.58$ cylinders will be required. So the final answer will be $84 + 1200 = 1284$.

One more fact to be noted is that the file occupies 83.58 cylinders, but the 0.58 cannot be accommodated in the first one (the files storage starts from < 1200, 9, 40 >)

Hence, the file will be extended to 194 855 – 85400) more bytes of cylinder 1284.

2. (b)

K-map for $F(P, Q, R, S) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$ where 2, 7, 8, 13 are do not care terms

RS	RS	RS	RS	RS
PO	1	1	1	1
PO	1	1	1	1
PO	1	1	1	1
PO	1	1	1	1

There are 2 quads.

$$\text{Quad 1 } (m_0 + m_2 + m_8 + m_{10}) = \bar{Q}\bar{S}$$

$$\text{Quad 2 } (m_5 + m_7 + m_{13} + m_{15}) = QS$$

Hence, minimal SOP form = $QS + \bar{Q}\bar{S}$

3. (c)

The given Venn diagram represent XNOR expression.

Hence, a XNOR b = $ab + a'b'$

4. (b)

The range of values for n-bit signed magnitude representation is $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$.

5. (a)

We know that,

$$\text{Average latency} = \frac{\text{Max Latency}}{2}$$

$$\text{Where, Max latency} = \frac{1}{\text{rev/time}} = \frac{1}{\frac{6000}{60} \text{ S}}$$

$$= \frac{60}{6000} \text{ S} = \frac{1}{100} \text{ S}$$

$$\text{So the average latency} = \frac{10^{-2}}{2}$$

$$= 0.5 \times 10^{-2}$$

$$= 5 \times 10^{-3} \text{ s}$$

6. (d)

8 bit representation of 100 $\rightarrow 01100100$

Complement $\rightarrow 10011011$

Add 1 $\rightarrow 00000001$

$\rightarrow 10011100$

Hence 2's complement representation is 10011100.





7. (*)

None option is correct.

8. (b)

9. (c)

Given, $(43)_a = (y3)_b$

By converting to decimal system

$$4x + 3x^0 = y. 8^1 + 3.8^0$$

$$4x + 3 = 8y + 3$$

$$4x = 8y$$

$$x = 2y$$

x is the base of number system $x > 4$.

Y is number in the number system with base 8, so $y < 8$.

x	y
6	3
8	4
10	5
12	6
14	7

Number of possible solution = 5

10. (d)

$$(1101)_2 - (1010)_2$$

1's complement of 1010 is 0101.

Minued	-1101
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1's complement of subtrahend -0101

Carry over	10010
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$$\begin{array}{r} 1 \\ \hline 0011 \end{array}$$

The required difference is 0011.

